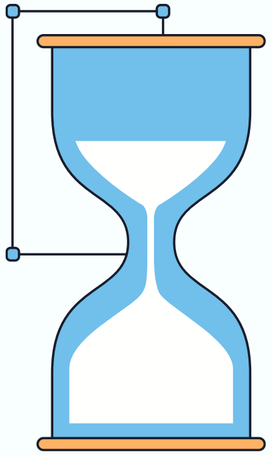
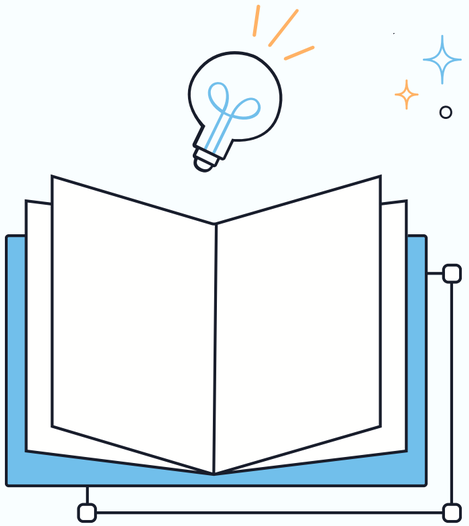
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Motorcycle Maneuvering in SUMO

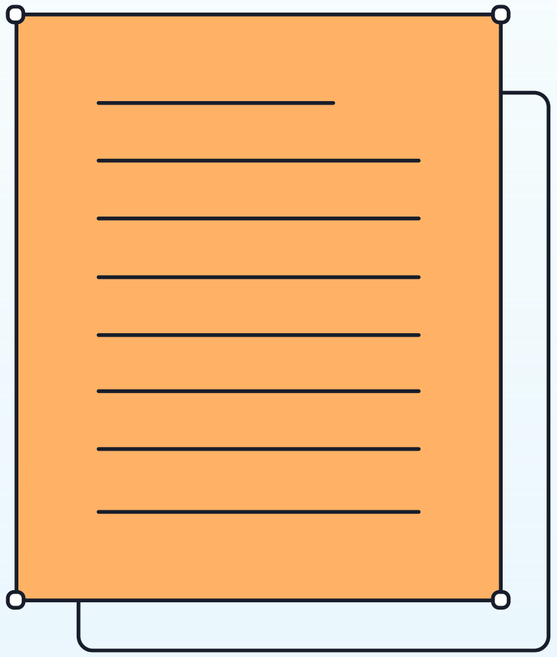
Speed-Gain Motivation, Lane Filtering, Lateral Interaction, and Surrogate Safety Measures

Using DataFromSky Trajectories for Model Development and Validation



Prepared by:
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Date
October 13, 2026



Presentation Overview

- Four Key Components of Motorcycle Behavior
- Lane-Changing Motivation in SUMO
- Lane Filtering vs. Conventional Lane Changing
- Lateral Conflicts in SUMO
- Surrogate Safety Measures
- Integration Workflow: DataFromSky + SUMO
- Validation Framework
- Recommendations



Four Distinct Components

Component	Question
1. Lane-Changing Motivation	Why does a motorcycle decide to move laterally?
2. Lateral Maneuver Execution	How does the motorcycle move into a gap or pass another vehicle?
3. Lateral Interaction & Conflict	How does the motorcycle respond to nearby vehicles while manoeuvring?
4. Surrogate Safety Measurement	How do we quantify the safety of these maneuvers?

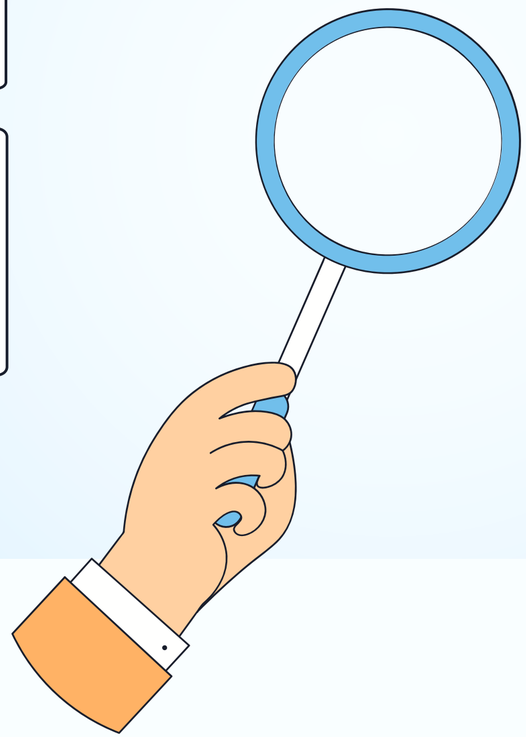




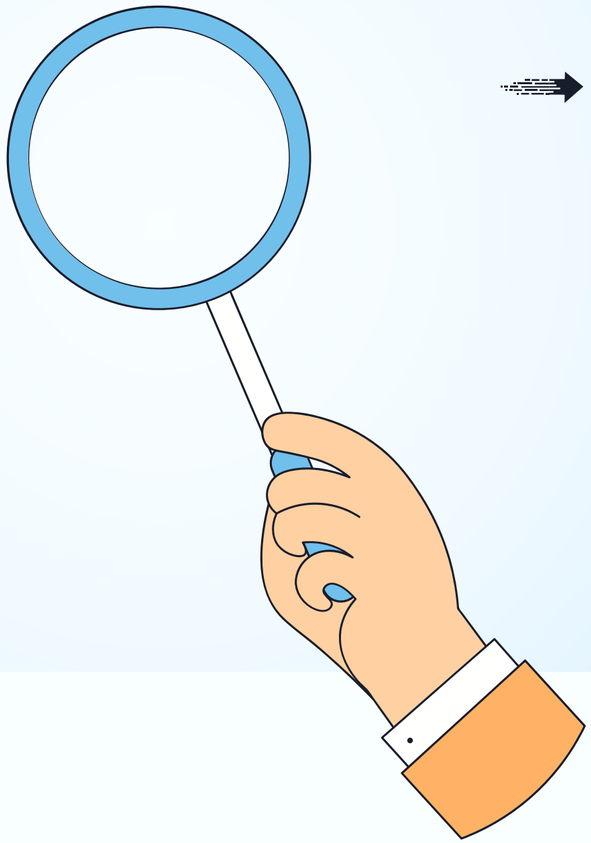
⚠️ These components are **related but not interchangeable**

Why This Distinction Matters

- A vehicle can change lanes due to speed advantage without explicitly evaluating lateral collision risk
- A simulation can reproduce lateral movement without measuring whether that movement creates danger
- Research gap: SUMO's baseline model may not capture motorcycle-specific filtering decisions, and standard SSM device excludes lateral conflicts



SUMO Lane-Changing Models



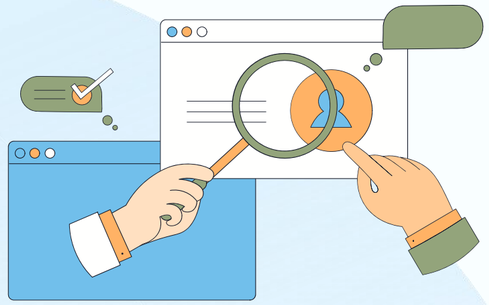
Available Models:

1. Default lane-changing model
2. SL2015 model (with sublane simulation)

Key Parameters:

Parameter	Function
lcSpeedGain	Eagerness to change lanes for speed gain
lcSpeedGainLookahead	Look-ahead horizon for speed-gain considerations
lcStrategic	Strategic lane change motivation
lcCooperative	Cooperative lane change behavior
lcKeepRight	Lane-keeping obligation
lcAssertive	Aggressiveness in lane-changing decisions

Speed-Gain Motivation (Conceptual Model)



Scenario: Motorcycle approaches signalized intersection in congested stream

Potential situations:

- ✓ Slowly moving car ahead
- ✓ Gap between two cars
- ✓ Adjacent lane with shorter queue
- ✓ Opportunity to travel beside a car
- ✓ Opportunity to filter forward



Conceptual representation:

$$M_{speed} = f(v_{target} - v_{current}, \text{traffic conditions}, \text{look-ahead}, \text{parameters})$$

⚠ This is conceptual—not exact SUMO implementation

Understanding lcSpeedGain

What it does:

- ✓ Controls how strongly a vehicle responds to potential speed advantages
- ✓ Not a complete model of motorcycle lane filtering

Example:
Motorcycle behind car at 8 km/h
Adjacent lane offers 15 km/h
Potential speed advantage: Δv = 7 km/h

What it does NOT evaluate:

- ✓ Lateral distance to car
- ✓ Gap width between vehicles
- ✓ Lateral closing speed
- ✓ Risk of handlebar/body collision
- ✓ Probability of successful passage



Understanding lcSpeedGainLookahead

Function:

- ✓ Controls look-ahead horizon for speed-gain considerations
- ✓ Determines how far ahead vehicle evaluates potential speed advantages

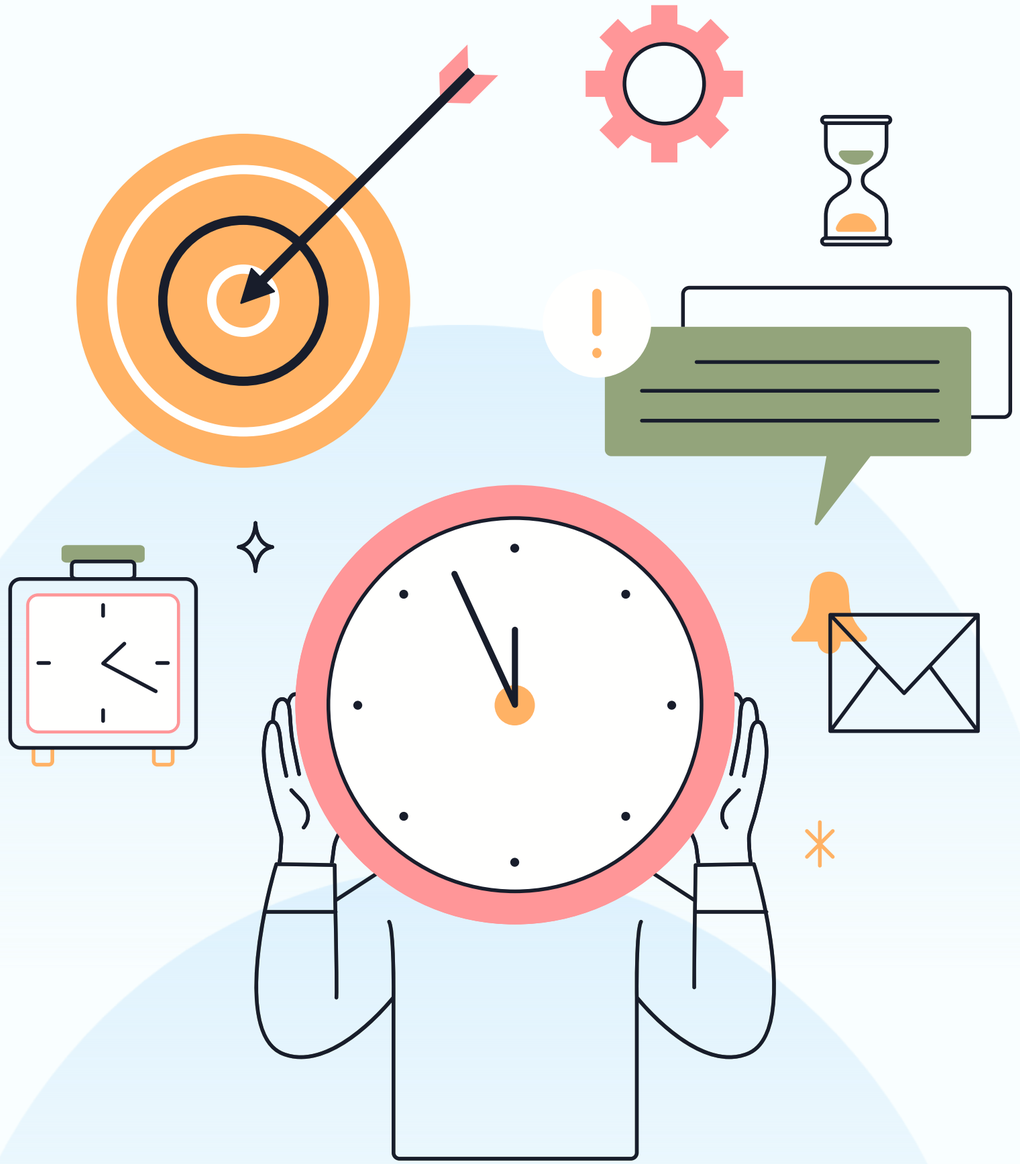
Importance:

- ✓ Motorcycle approaching queue several vehicle lengths ahead
- ✓ Decision based only on immediate vehicle ≠ decision considering downstream conditions

⚠️ **Critical distinction:** Looking farther ahead for speed advantage ≠ evaluating lateral safety



Motivation Accumulation - Technical Clarification



SUMO's lane-changing models evaluate internal motivations and constraints

SL2015 model considers:

- ✓ Speed gain
- ✓ Strategic lane changes
- ✓ Cooperation
- ✓ Lane-keeping obligations
- ✓ Lateral alignment (when sublane enabled)
- ✓ Gap safety constraints

NOT a simple universal rule:

"Change lane if accumulated motivation"> "threshold"

lcSpeedGain ≠ universal motivation threshold

lcSpeedGainLookahead ≠ motivation accumulation time

Decision Process Example

Motorcycle approaching signalized intersection:

- ✓ Follows slow car
- ✓ Identifies potential speed advantage beside car
- ✓ Develops motivation to change lateral position
- ✓ Evaluates whether maneuver is permitted by lane-changing model
- ✓ Begins maneuver if model accepts request

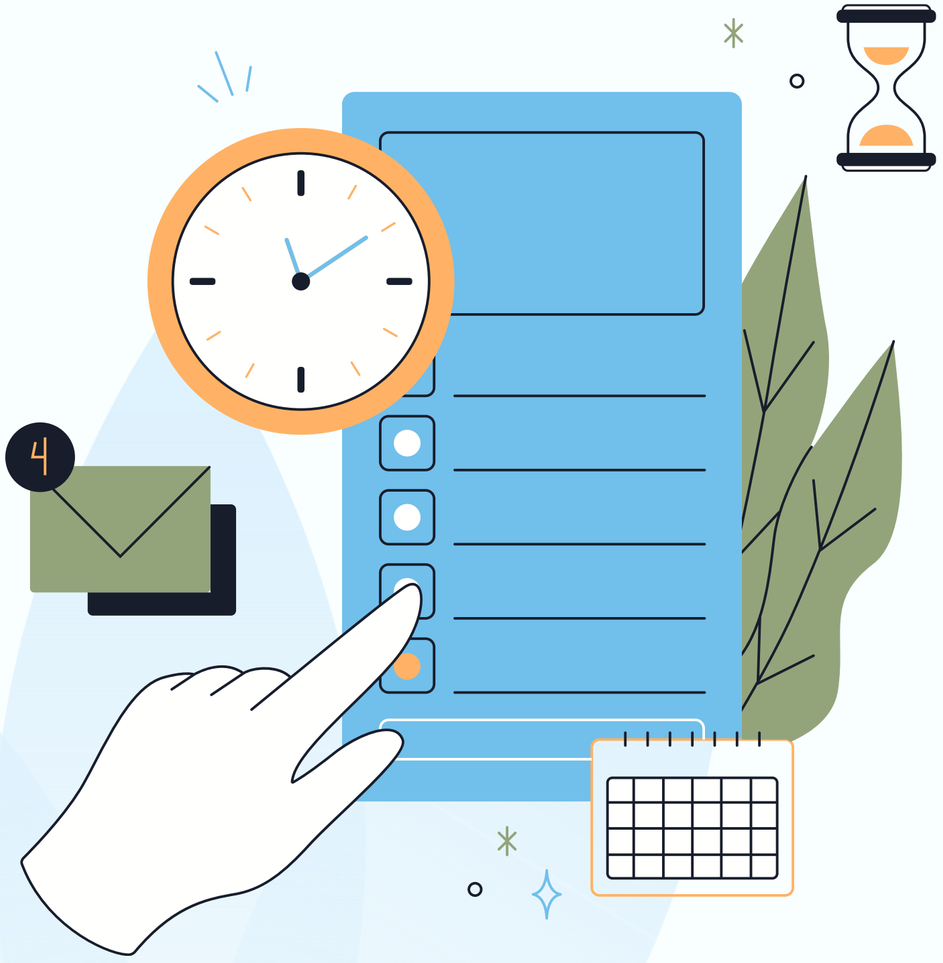
Key research question:

Does the model capture actual decision-making of motorcycle filtering through queue?



Lane Filtering vs. Conventional Lane Changing

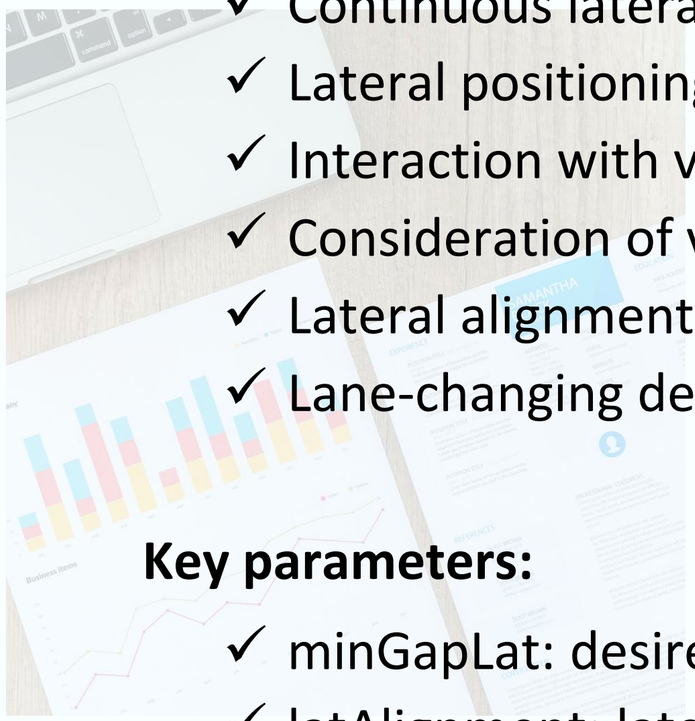
Feature	Conventional Lane Changing	Motorcycle Lane Filtering
Primary movement	From one lane to another	Through lateral gap, potentially same nominal lane
Typical motivation	Speed gain, route requirements	Progress through congestion, queue advancement
Main geometric constraint	Target-lane availability, longitudinal gaps	Available width between vehicles, lateral clearance
Key behavioral variables	Speed advantage, gap acceptance, urgency	Gap acceptance, lateral position, motorcycle width, relative motion
Safety concern	Conflict with target lane vehicles	Side-swipe, lateral encroachment, collision with passed vehicles



SUMO's Sublane Model

What it enables:

- ✓ Continuous lateral movement
- ✓ Lateral positioning within a lane
- ✓ Interaction with vehicles in neighboring lateral positions
- ✓ Consideration of vehicle widths and lateral gaps
- ✓ Lateral alignment preferences
- ✓ Lane-changing decisions using SL2015 model



Key parameters:

- ✓ minGapLat: desired minimum lateral gap
- ✓ latAlignment: lateral positioning preferences
- ✓ Lateral resolution: discretization of road width



Sublane Model Limitations

Critical limitation:

Representing lateral movement ≠ modelling full behavioral process that produces motorcycle lane filtering

Built-in model can represent aspects of lateral interaction, but does NOT automatically reproduce:

- ✓ Every motorcycle-specific decision rule
- ✓ Individual differences in gap acceptance
- ✓ Speed-dependent behavior variations
- ✓ Urgency effects on decision-making



What to Calibrate from DataFromSky

Component	What to Estimate
Speed-gain motivation	Whether motorcycles move laterally when adjacent space offers speed advantage
Gap acceptance	Minimum and preferred longitudinal and lateral gaps
Lateral movement	Lateral speed, acceleration, maneuver duration
Lateral positioning	Position relative to lane centre and neighboring vehicles
Filtering probability	Probability under different queue lengths, speeds, gap widths
Safety response	Whether motorcycles slow down, change direction, or abandon maneuver
Queue advancement	How motorcycles approach and position near stop line



Lateral Interactions in SUMO Simulation

When sublane model is enabled, SUMO accounts for:

- ✓ Lateral positioning
- ✓ Lateral gaps between vehicles
- ✓ Prevention of incompatible lateral positions

Relevant parameters:

Parameter	Function
minGapLat	Desired minimum lateral gap
lcPushy	Aggressive lateral encroachment
lcImpatience	Acceptance of smaller gaps when blocked
lcSigma	Random lateral variations
lcAccelLat	Lateral acceleration limits



Lateral Gap Constraints ≠ Conflict Risk Model

Important distinction:
Maintaining specified lateral gap at simulation step:

- ✓ Does NOT quantify collision probability over next second
- ✓ Does NOT determine if trajectories will intersect
- ✓ Does NOT guarantee all realistic filtering maneuvers reproduced

It would be incorrect to say SUMO entirely ignores lateral interaction



Standard SUMO SSM Device

Measures calculated:

Measure	Description
TTC	Time to Collision
DRAC	Deceleration Rate to Avoid a Crash
PET	Post-Encroachment Time
MDRAC	Modified Deceleration Rate to Avoid a Crash
Additional	Brake rate, spacing gap, time gap

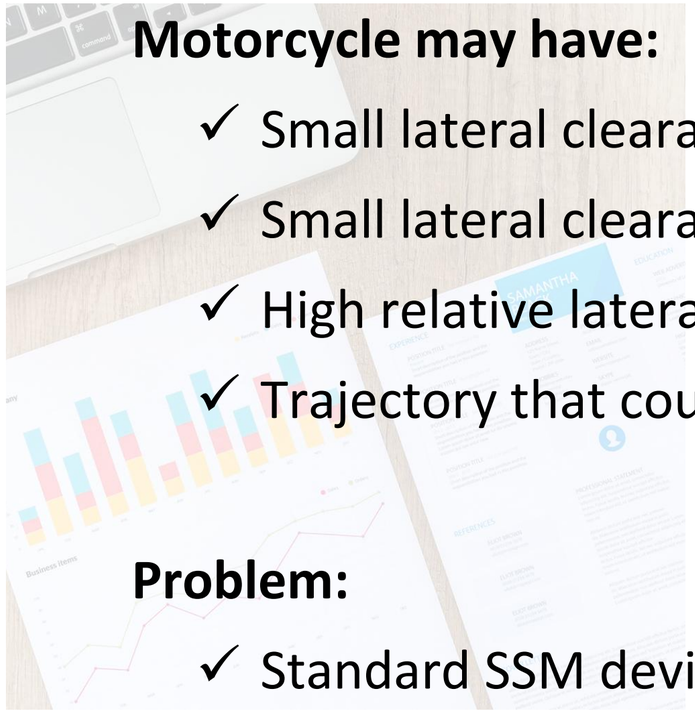
⚠️ **Critical documented limitation:**

SUMO's SSM device **ignores lateral conflicts** involving traffic in opposite or neighboring lanes



The Lateral Conflict Problem

Scenario: Motorcycle passing between two stationary cars at signalized intersection



Motorcycle may have:

- ✓ Small lateral clearance from car on left
- ✓ Small lateral clearance from car on right
- ✓ High relative lateral closing speed
- ✓ Trajectory that could result in side-swipe collision

Problem:

- ✓ Standard SSM device does not provide complete evaluation
- ✓ Simulation might generate important filtering maneuver
- ✓ Standard SSM output might not identify lateral conflict

⚠️ This is a **measurement limitation**, not necessarily a simulation limitation

Why Ordinary TTC is Insufficient

Two vehicles approaching laterally:

- d_{lat} = lateral clearance
- $v_{lat,rel}$ = relative lateral closing speed

Simple lateral time-to-contact:

$$TTC_{lat} = \frac{d_{lat}}{v_{lat,rel}}$$

Example:

- Lateral clearance: 0.30 m
- Relative lateral speed: 0.60 m/s
- $TTC_{lat} = 0.50$ s → potentially critical



⚠️ **Limitation:** Does not establish collision will occur

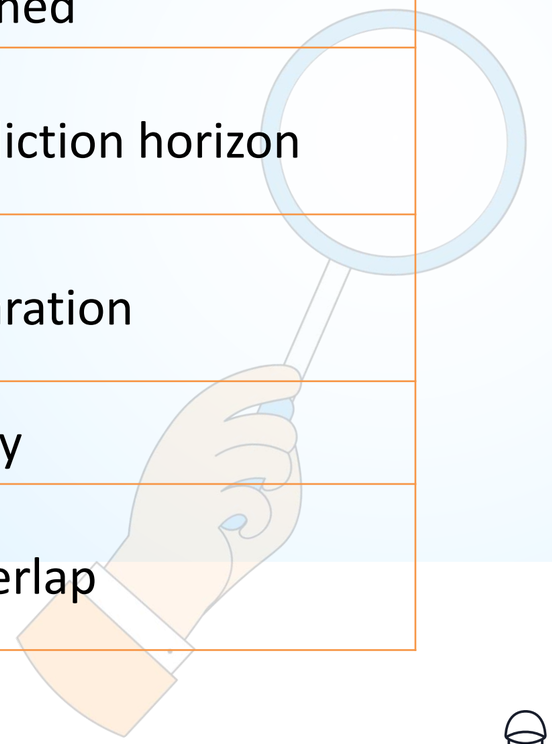
- Must consider longitudinal separation, vehicle dimensions, orientations, future trajectories
- Lateral TTC from lateral clearance alone ≠ complete two-dimensional TTC

Supplementary Lateral Safety Measures

Strategy: Retain existing SSM measures + develop supplementary lateral safety analysis

Proposed measures:

Measure	Purpose
Lateral TTC	Time until critical lateral clearance reached
Minimum Predicted Distance (MPD)	Smallest expected separation over prediction horizon
Time to Closest Approach (TCA)	When trajectories reach minimum separation
PET for Lateral Maneuvers	Time between conflict region occupancy
2D TTC	Time until predicted occupied areas overlap



Lateral TTC (TTC_lat)

Definition:

$$TTC_{lat} = \begin{cases} \frac{d_{lat}}{v_{lat,rel}} & v_{lat,rel} > 0 \\ \infty & v_{lat,rel} \leq 0 \end{cases}$$

Application:

- ✓ Identify filtering maneuvers where lateral clearance decreases rapidly

Important limitation:

- ✓ Does not determine if vehicles occupy same longitudinal position at predicted contact time
- ✓ Combine with longitudinal separation or 2D collision prediction



Minimum Predicted Distance (MPD)

Definition:

Predicted positions:

$$\mathbf{p}_1(t) = \begin{bmatrix} x_1(t) \\ y_1(t) \end{bmatrix}, \quad \mathbf{p}_2(t) = \begin{bmatrix} x_2(t) \\ y_2(t) \end{bmatrix}$$

Centre-to-centre separation:

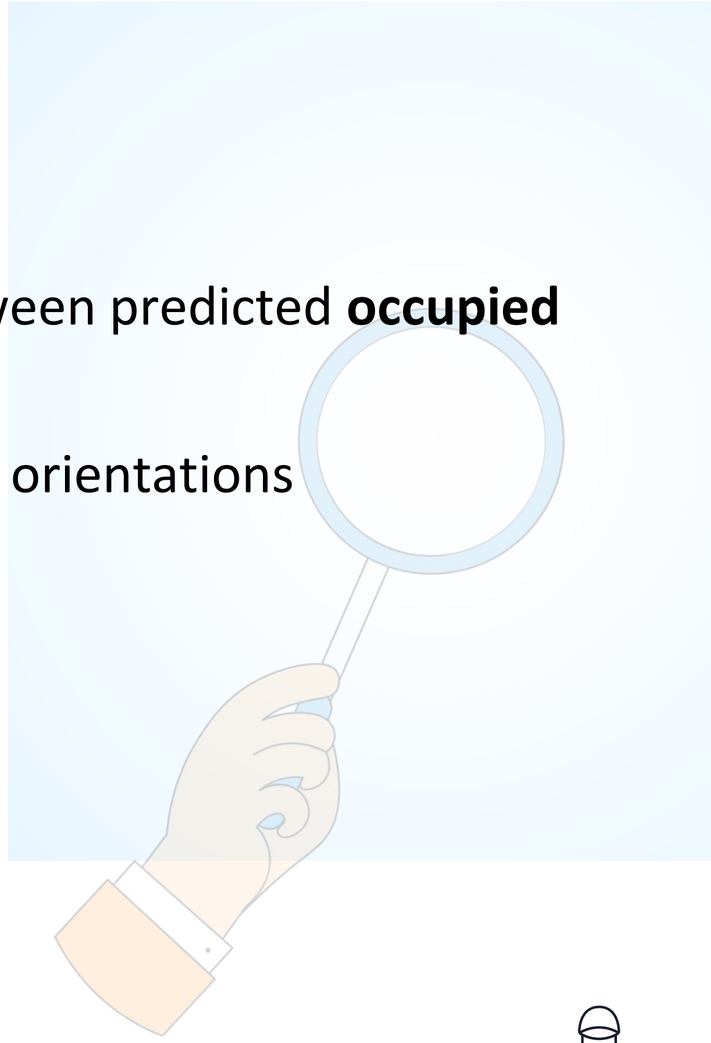
$$D(t) = \|\mathbf{p}_1(t) - \mathbf{p}_2(t)\|$$

Minimum predicted distance:

$$MPD_H = \min_{0 \le t \le H} D(t)$$

For motorcycle application:

- Use minimum distance between predicted **occupied areas** (not just centers)
- Account for lengths, widths, orientations



Time to Closest Approach (TCA)

For constant relative velocity:

$$\mathbf{r}_0 = \mathbf{p}_1(0) - \mathbf{p}_2(0)$$
$$\mathbf{v}_{rel} = \mathbf{v}_1 - \mathbf{v}_2$$

Time to closest approach:

$$TCA = -\frac{\mathbf{r}_0 \cdot \mathbf{v}_{rel}}{\|\mathbf{v}_{rel}\|^2}$$

Application:

- Detect situations where motorcycle moves forward while moving toward adjacent car

⚠️ **Note:** TCA is not itself a collision indicator

- More meaningful when combined with predicted minimum separation



Two-Dimensional TTC (TTC_2D)

Most promising direction for research

Process at each time step:

1. Obtain motorcycle position, speed, heading, dimensions
2. Obtain corresponding information for neighboring vehicles
3. Predict future trajectories
4. Represent each vehicle as 2D shape
5. Determine if predicted occupied areas overlap
6. Calculate earliest predicted time of overlap

Definition:

$$TTC_{2D} = \inf\{t > 0 : A_{moto}(t) \cap A_{other}(t) \neq \emptyset\}$$

Where $A_{moto}(t)$ and $A_{other}(t)$ are predicted occupied areas



TTC_2D Advantages

Captures both dimensions simultaneously:

✅ Vehicles may be close laterally but separated longitudinally → no collision

✅ Adequate lateral clearance now but trajectories will intersect → collision predicted

Research advantage:

- Evaluates joint effect of longitudinal and lateral movement
- Particularly relevant to motorcycle filtering

⚠️ **Important:**

- TTC_2D is a **family of implementations**, not one standardized formula
- Must specify: trajectory prediction, vehicle geometry, collision criterion, uncertainty treatment



Integration Workflow Overview

Three-Part Workflow:

Part A: Extract Motorcycle Manoeuvres from DataFromSky

↓

Part B: Develop and Calibrate SUMO Motorcycle Model

↓

Part C: Calculate Supplementary Safety Measures



Part A - Extract from DataFromSky

Extract for every motorcycle:

Variable	Purpose
(x,y) coordinates	Reconstruct 2D movement
Longitudinal position	Progress towards stop line
Lateral position	Quantify filtering and lateral displacement
Longitudinal speed	Measure speed gain
Lateral speed	Quantify lateral movement
Longitudinal acceleration	Characterize braking/acceleration
Lateral acceleration	Characterize lateral manoeuvres
Motorcycle width/length	Estimate occupied space
Neighboring vehicle positions	Identify potential interactions
Longitudinal/lateral gaps	Characterize manoeuvre feasibility
Signal state, distance to stop line	Explain intersection approach influence



Deriving Lateral Speed and Acceleration

Lateral speed (finite difference):

$$v_y(t) \approx \frac{y(t+\Delta t)-y(t-\Delta t)}{2\Delta t}$$

Lateral acceleration (finite difference):

$$a_y(t) \approx \frac{y(t+\Delta t)-2y(t)+y(t-\Delta t)}{(\Delta t)^2}$$

⚠️ **Trajectory smoothing may be necessary** — measurement noise strongly affects acceleration estimates



Behavior Classification

Classify motorcycle behavior into categories:

- ✓ Following a car
- ✓ Filtering between two cars
- ✓ Filtering beside a queued vehicle
- ✓ Moving towards stop line
- ✓ Changing lateral position to avoid vehicle
- ✓ Aborting a filtering maneuver
- ✓ Remaining stationary in queue

⚠️ **Important:** Not every lateral movement is a lane-filtering maneuver



Part B - Two Simulation Configurations

Model 1: Baseline SUMO Model

- ✓ Sublane model + SL2015 lane-changing model
- ✓ Suitable motorcycle dimensions and behavioral parameters
- ✓ Calibrate: lcSpeedGain, lcSpeedGainLookahead, minGapLat, lcPushy, lcImpatience, lcSigma, lateral speed/acceleration settings

Model 2: Enhanced Motorcycle Filtering Model

- ✓ Additional behavioral component using observed trajectories
- ✓ Represents motorcycle-specific filtering decisions



Enhanced Filtering Decision Model

Proposed structure (not established equation):

$$P(\text{filter}) = f(w_{gap}, g_{long}, g_{lat}, \Delta v, d_{stop}, q, s)$$

Where:

- w_{gap} = available lateral gap width
- g_{long} = available longitudinal gap
- g_{lat} = current lateral clearance
- Δv = potential speed advantage
- d_{stop} = distance to stop line
- q = queue conditions
- s = signal state

Starting point: Logistic regression or other probabilistic model



Staged Maneuver Model

Model filtering in stages:

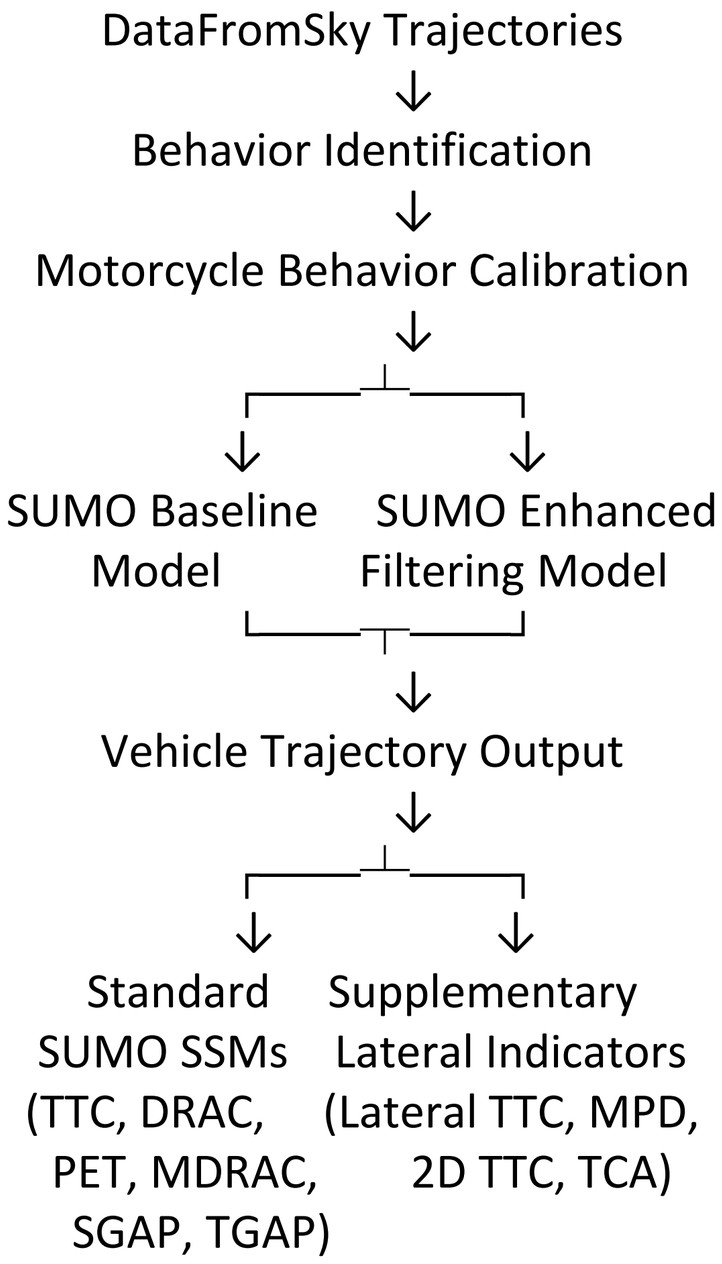
1. **Decision:** Does motorcycle initiate filtering?
2. **Gap selection:** Which available gap does it select?
3. **Lateral execution:** How does it move through gap?
4. **Interaction response:** Does it slow down, adjust lateral position, or abandon?

Why this separation is valuable:

- Decision to filter ≠ physical execution of filtering
- Different processes require different modeling approaches



Part C - Practical Architecture



Validation Framework

1 Behavioral Validation

Compare simulated and observed:

- Number of filtering maneuvers
- Gap-acceptance distributions
- Motorcycle positioning within queues
- Speed profiles during filtering

2 Geometric and Interaction Validation

Compare:

- Minimum lateral clearance
- Minimum longitudinal clearance
- Minimum 2D clearance
- Frequency of very small gaps
- Relative speed during filtering
- Abrupt lateral maneuvers
- Overlap/near-overlap events



Safety Validation Comparison

Safety Measure	Baseline SUMO	Enhanced Model	Research Question
TTC	Standard output	Standard output	Does longitudinal/intersection-conflict risk change?
DRAC	Standard output	Standard output	Do required evasive decelerations differ?
PET	Standard output	Standard output	Does post-encroachment risk change?
Lateral TTC	Custom calculation	Custom calculation	Does lateral conflict risk differ?
MPD	Custom calculation	Custom calculation	Does minimum predicted separation change?
2D TTC	Custom calculation	Custom calculation	Does 2D collision risk differ?



Defensible Research Gap

- ✓ Although SUMO's sublane model can represent lateral vehicle movement and lateral-gap interactions, its standard lane-changing behavior does not necessarily reproduce motorcycle-specific lane-filtering decisions.

- ✓ Furthermore, the standard SSM device explicitly excludes lateral conflicts involving neighboring or opposing traffic. Consequently, an enhanced motorcycle behavior model combined with supplementary two-dimensional safety indicators may be needed to evaluate motorcycle filtering at signalized intersection approaches adequately.



Two Separate Problems

1. Behavioral Limitation

- Baseline model may not reproduce observed decision-making
- May not match actual movement patterns of motorcycles

2. Safety-Evaluation Limitation

- Standard SSM device does not comprehensively evaluate lateral conflicts
- Important maneuvers may go unmeasured

Can investigate both using same trajectory dataset — but require different validation procedures



Recommendations

1. **Characterize** motorcycle filtering using DataFromSky trajectories
2. **Calibrate** baseline SUMO sublane model to establish how well existing mechanisms reproduce observed behavior
3. **Develop** motorcycle-specific filtering model only for behaviors baseline fails to reproduce
4. **Develop** supplementary 2D safety-analysis module (lateral clearance, lateral TTC, MPD)
5. **Validate** behavior and safety indicators independently against observed trajectories
6. **Compare** baseline and enhanced models under different traffic volumes, queue lengths, signal conditions



Practical Starting Point

Begin with these three indicators:

Indicator	Description
Minimum lateral clearance	Smallest lateral gap during manoeuvre
Lateral TTC	Time until critical lateral clearance
Minimum predicted 2D distance	Smallest predicted separation between occupied areas

These three are relatively complementary

Once validated:

- Develop more sophisticated 2D TTC based on vehicle geometry and predicted trajectories



Key Takeaways

1. **Four distinct components** — motivation, execution, interaction, safety measurement
2. lcSpeedGain **≠ motorcycle filtering model** — it's one parameter among many
3. **Sublane model enables lateral movement** but doesn't guarantee realistic filtering behavior
4. **Standard SSM device excludes lateral conflicts** — critical limitation for motorcycle research
5. **Supplementary lateral safety measures** needed: Lateral TTC, MPD, 2D TTC
6. **Three-part workflow:** Extract → Calibrate → Calculate
7. **Validate behavior and safety independently**
8. **Start simple:** minimum lateral clearance, lateral TTC, MPD



Thank You

